What each operator makes short

Shortest construction of calculator functions, and how often GP found them

	EML	EDL	−EML	LDE_0	LDE_1	PLI	PLM
ONE VARIABLE							
e^x	**3**	**3**	7	7	9	7	15
ln x	7	7	**3**	**3**	5	**3**	19
−x	**15**	–	17	17	**15**	–	–
1/x	**15**	17	–	17	**15**	17	17
x^2	**17**	19	–	19	–	**17**	–
x + 1	**19**	–	23	23	23	–	–
x − 1	**11**	**11**	15	19	23	17	–
2x	19	–	19	**17**	23	–	–
e^{-x}	17	–	23	15	**11**	23	23
x^x	–	19	–	–	–	21	**9**
$\exp(e^x)$	**5**	**5**	–	–	–	–	–
ln(ln x)	13	–	**5**	**5**	9	**5**	–
TWO VARIABLES							
x + y	–	–	–	**17**	–	–	–
x − y	**11**	–	**11**	17	–	–	–
x · y	**17**	–	–	–	–	**17**	**17**
x / y	17	**11**	13	**11**	15	17	17
x^y	–	–	–	–	–	–	**9**
$\log_x y$	–	15	17	**7**	11	13	13

Share of 30 GP runs that found the function (2×10^4 evaluations)

0% 50% 100%

Numbers: shortest construction in nodes (bold: shortest across operators). Outlined: not searched (fails the viability filter).